1)

$y' = \frac{2xy}{x^2-1}$ ~~[illegible]~~ ~~[illegible]~~ $x^2-1=0$

$|x|=1$

$y' = \frac{2x}{x^2-1} \cdot y$

$\frac{y'}{y} = \frac{2x}{x^2-1}$ → stac. řešení $y=0$

$\int \frac{dy}{y} = \int \frac{2x}{x^2-1}\,dx$ ~~→ stac. řešení~~

$\ln|y| = \ln|x^2-1| + C$

$e^{\ln|y|} = e^{\ln|x^2-1|+C}$

$\pm y = e^{C} \cdot (\pm x^2 - 1)$ $\quad 0\cdot(x^2-1) = 0 \Rightarrow y = D\cdot(x^2-1)$ zahrnuje $y=0$

$\boxed{y = D\cdot(x^2-1),\ x \neq \pm 1}$

a)

$y(0)=1$

$1 = D\cdot(0-1)$

$1 = D\cdot(-1)$

$D = (-1)$

$\boxed{y = 1-x^2}$

b)

$y(2) = (-3)$

$-3 = D\cdot(4-1)$

$\frac{-3}{3} = D$

$D = -1$

$\boxed{y = 1-x^2}$

c)

$y(-3) = 16$

$16 = D\cdot(9-1)$

$\frac{16}{8} = D$

$D = 2$

$\boxed{y = 2\cdot(x^2-1)}$

d)

$y(2) = 0$

$0 = D\cdot(4-1)$

$D = 0$

$\boxed{y = 0}$

2)

$$y' - 2xy = e^{x^2}$$

$$e^{-x^2} y' - 2e^{-x^2} xy = 1$$

$$e^{-x^2} y' - (e^{-x^2})' y = 1$$

$$e^{-x^2} \frac{dy}{dx} - \frac{d}{dx}(e^{-x^2}) y = 1$$

$$\cancel{\frac{d}{dx}}(e^{-x^2} y)' = 1$$

$$\int (e^{-x^2} y)' \, dx = \int 1 \, dx$$

$$e^{-x^2} y = x + C$$

$$\boxed{y = e^{x^2}(x + C)}$$